

EECS 16A Foundations of Signals, Systems, & Information Processing

Fall 2025 Homework 6

This homework is due Friday, October 10, 2025, at 23:59.

Submission Format

Your homework submission should consist of a single PDF file that contains all of your answers (any handwritten answers should be scanned).

On occasion, a problem set contains one more problems designated as optional. Your solutions to such problems are not to be evaluated, and will not count toward your grade. Nevertheless, you are still responsible for learning the subject matter within their scope. In other words, we may test you on their scope.

1. Continuous-Time Complex Exponential Projections

Consider the space of continuous-time complex-valued signals over one period, say, over the interval $[0, p]$, where $p > 0$. Define the inner product between two signals x and y as:

$$\langle x, y \rangle = \frac{1}{p} \int_0^p x(t) y^*(t) dt$$

and the induced norm: $\|x\| = \sqrt{\langle x, x \rangle}$.

Let $\phi_k(t) = e^{ik\omega_0 t}$, where $\omega_0 = \frac{2\pi}{p}$, for integer k .

Let $x(t) = t$ for $t \in [0, p]$.

- (a) Find the projection of x onto ϕ_0 , i.e., find $\alpha_0 = \langle x, \phi_0 \rangle$.

Solution:

$$\alpha_0 = \langle t, 1 \rangle = \frac{1}{p} \int_0^p t \cdot 1 dt = \frac{1}{p} \cdot \frac{p^2}{2} = \frac{p}{2}.$$

- (b) Find the projection of x onto ϕ_1 , i.e., α_1 .

Solution:

$$\alpha_1 = \langle t, e^{i\omega_0 t} \rangle = \frac{1}{p} \int_0^p t e^{-i\omega_0 t} dt.$$

Integrate by parts: let $u = t$, $dv = e^{-i\omega_0 t} dt$, so $du = dt$, $v = \frac{e^{-i\omega_0 t}}{-i\omega_0}$.

$$\int_0^p t e^{-i\omega_0 t} dt = \left[t \cdot \frac{e^{-i\omega_0 t}}{-i\omega_0} \right]_0^p - \int_0^p \frac{e^{-i\omega_0 t}}{-i\omega_0} dt = \frac{p e^{-i2\pi}}{-i\omega_0} - 0 + \frac{1}{i\omega_0} \int_0^p e^{-i\omega_0 t} dt.$$

Since $e^{-i2\pi} = 1$, and $\int_0^p e^{-i\omega_0 t} dt = 0$ (orthogonality for $k = 1$), we get:

$$\int_0^p t e^{-i\omega_0 t} dt = \frac{p}{-i\omega_0} = \frac{ip}{\omega_0}.$$

Thus,

$$\alpha_1 = \frac{1}{p} \cdot \frac{ip}{\omega_0} = \frac{i}{\omega_0}.$$

(c) Find α_{-1} .

Solution:

$$\alpha_{-1} = \langle t, e^{-i\omega_0 t} \rangle = \frac{1}{p} \int_0^p t e^{i\omega_0 t} dt.$$

Similar to above, integrate by parts:

$$\int_0^p t e^{i\omega_0 t} dt = \left[t \cdot \frac{e^{i\omega_0 t}}{i\omega_0} \right]_0^p - \int_0^p \frac{e^{i\omega_0 t}}{i\omega_0} dt = \frac{p e^{i2\pi}}{i\omega_0} - 0 - \frac{1}{i\omega_0} \cdot 0 = \frac{p}{i\omega_0} = -\frac{ip}{\omega_0}.$$

$$\text{So } \alpha_{-1} = \frac{1}{p} \cdot \left(-\frac{ip}{\omega_0} \right) = -\frac{i}{\omega_0}.$$

(d) Using your answers above, form the approximation $\hat{x}(t) = \alpha_0 \phi_0(t) + \alpha_1 \phi_1(t) + \alpha_{-1} \phi_{-1}(t)$.

Solution:

$$\hat{x}(t) = \frac{p}{2} \phi_0(t) + \frac{i}{\omega_0} \phi_1(t) - \frac{i}{\omega_0} \phi_{-1}(t)$$

(e) Write $\hat{x}(t)$ in terms of real sinusoids.

Solution:

$$\hat{x}(t) = \frac{p}{2} \cdot 1 + \frac{i}{\omega_0} e^{i\omega_0 t} - \frac{i}{\omega_0} e^{-i\omega_0 t}.$$

Note: $\frac{i}{\omega_0} e^{i\omega_0 t} - \frac{i}{\omega_0} e^{-i\omega_0 t} = \frac{i}{\omega_0} \cdot 2i \sin(\omega_0 t) = -\frac{2}{\omega_0} \sin(\omega_0 t)$. So:

$$\hat{x}(t) = \frac{p}{2} - \frac{2}{\omega_0} \sin(\omega_0 t).$$

(f) Compute the approximation error energy: $\|x - \hat{x}\|^2$.

Solution: By orthogonality and properties of projections:

$$\|x - \hat{x}\|^2 = \|x\|^2 - \|\hat{x}\|^2 = \langle x, x \rangle - (|\alpha_0|^2 + |\alpha_1|^2 + |\alpha_{-1}|^2).$$

Compute:

$$\|x\|^2 = \frac{1}{p} \int_0^p t^2 dt = \frac{1}{p} \cdot \frac{p^3}{3} = \frac{p^2}{3}.$$

$$|\alpha_0|^2 = \left(\frac{p}{2} \right)^2 = \frac{p^2}{4}, \quad |\alpha_1|^2 = |\alpha_{-1}|^2 = \left| \frac{i}{\omega_0} \right|^2 = \frac{1}{\omega_0^2} = \frac{p^2}{4\pi^2}.$$

So:

$$\|\hat{x}\|^2 = \frac{p^2}{4} + 2 \cdot \frac{p^2}{4\pi^2} = \frac{p^2}{4} \left(1 + \frac{2}{\pi^2} \right).$$

Thus:

$$\|x - \hat{x}\|^2 = \frac{p^2}{3} - \frac{p^2}{4} \left(1 + \frac{2}{\pi^2} \right) = p^2 \left(\frac{1}{3} - \frac{1}{4} - \frac{1}{2\pi^2} \right) = p^2 \left(\frac{1}{12} - \frac{1}{2\pi^2} \right).$$

This is the energy of the component of x orthogonal to the span of $\{\phi_{-1}, \phi_0, \phi_1\}$, i.e., the error from approximating x (a CT ramp) with $\hat{x}$.

2. Mechanical Gram-Schmidt

For each of the following sets of vectors, use Gram-Schmidt to construct an orthonormal basis that shares their span.

$$(a) D = \left\{ \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Solution:

We start with the vectors of D as our basis, and we want to find an orthonormal basis for this same space using Gram-Schmidt. For notational convenience, we define:

$$D = \{\mathbf{d}_1, \mathbf{d}_2, \mathbf{d}_3\}, U = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} \quad (1)$$

We summarize the first few steps of the Gram-Schmidt algorithm as follows:

- i. $\mathbf{u}'_1 = \mathbf{d}_1; \quad \mathbf{u}_1 = \frac{\mathbf{u}'_1}{\|\mathbf{u}'_1\|}.$
- ii. $\mathbf{u}'_2 = \mathbf{d}_2 - \langle \mathbf{d}_2, \mathbf{u}_1 \rangle \mathbf{u}_1; \quad \mathbf{u}_2 = \frac{\mathbf{u}'_2}{\|\mathbf{u}'_2\|}.$
- iii. $\mathbf{u}'_3 = \mathbf{d}_3 - \langle \mathbf{d}_3, \mathbf{u}_1 \rangle \mathbf{u}_1 - \langle \mathbf{d}_3, \mathbf{u}_2 \rangle \mathbf{u}_2; \quad \mathbf{u}_3 = \frac{\mathbf{u}'_3}{\|\mathbf{u}'_3\|}.$

For the span of D , this is

$$\text{i. } \mathbf{u}'_1 = \mathbf{d}_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}. \text{ Therefore, } \mathbf{u}_1 = \frac{\mathbf{u}'_1}{\|\mathbf{u}'_1\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ -\frac{1}{\sqrt{2}} \end{bmatrix}.$$

ii.

$$\mathbf{u}'_2 = \mathbf{d}_2 - \langle \mathbf{d}_2, \mathbf{u}_1 \rangle \mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - \frac{1}{\sqrt{2}} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ -\frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ 1 \\ \frac{1}{2} \end{bmatrix} \Rightarrow \mathbf{u}_2 = \begin{bmatrix} \frac{1}{\sqrt{6}} \\ \frac{\sqrt{2}}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} \end{bmatrix} \quad (2)$$

iii.

$$\mathbf{u}'_3 = \mathbf{d}_3 - \langle \mathbf{d}_3, \mathbf{u}_1 \rangle \mathbf{u}_1 - \langle \mathbf{d}_3, \mathbf{u}_2 \rangle \mathbf{u}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} - \left(-\frac{1}{\sqrt{2}} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ -\frac{1}{\sqrt{2}} \end{bmatrix} \right) - \frac{1}{\sqrt{6}} \begin{bmatrix} \frac{1}{\sqrt{6}} \\ \frac{\sqrt{2}}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} \end{bmatrix} = \begin{bmatrix} \frac{1}{3} \\ -\frac{1}{3} \\ \frac{1}{3} \end{bmatrix} \Rightarrow \mathbf{u}_3 = \begin{bmatrix} \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix} \quad (3)$$

Thus, our new orthonormal basis U is given by

$$U = \left\{ \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ -\frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{6}} \\ \frac{\sqrt{2}}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix} \right\} \quad (4)$$

$$(b) V = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} \right\}$$

Solution:

We start with the vectors of V as our basis, and we want to find an orthonormal basis for this same

space using Gram-Schmidt. As with the previous subpart, we define the following for notational convenience:

$$V = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}, U = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\} \quad (5)$$

We summarize the first few steps of the Gram-Schmidt algorithm as follows:

- i. $\mathbf{u}'_1 = \mathbf{v}_1; \quad \mathbf{u}_1 = \frac{\mathbf{u}'_1}{\|\mathbf{u}'_1\|}.$
- ii. $\mathbf{u}'_2 = \mathbf{v}_2 - \langle \mathbf{v}_2, \mathbf{u}_1 \rangle \mathbf{u}_1; \quad \mathbf{u}_2 = \frac{\mathbf{u}'_2}{\|\mathbf{u}'_2\|}.$
- iii. $\mathbf{u}'_3 = \mathbf{v}_3 - \langle \mathbf{v}_3, \mathbf{u}_1 \rangle \mathbf{u}_1 - \langle \mathbf{v}_3, \mathbf{u}_2 \rangle \mathbf{u}_2; \quad \mathbf{u}_3 = \frac{\mathbf{u}'_3}{\|\mathbf{u}'_3\|}.$

For the span of V , this is

- i. $\mathbf{u}'_1 = \mathbf{v}_1 = [1 \ 0 \ 0 \ 0 \ 0]^\top$. Since $\mathbf{u}'_1$ is already normalized, we simply set $\mathbf{u}_1 = \mathbf{u}'_1$.

ii.

$$\mathbf{u}'_2 = \mathbf{v}_2 - \langle \mathbf{v}_2, \mathbf{u}_1 \rangle \mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} \Rightarrow \mathbf{u}_2 = \begin{bmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \\ 0 \end{bmatrix} \quad (6)$$

iii.

$$\mathbf{u}'_3 = \mathbf{v}_3 - \langle \mathbf{v}_3, \mathbf{u}_1 \rangle \mathbf{u}_1 - \langle \mathbf{v}_3, \mathbf{u}_2 \rangle \mathbf{u}_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} - \frac{2}{\sqrt{2}} \begin{bmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \Rightarrow \mathbf{u}_3 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \quad (7)$$

Thus, our new orthonormal basis U is given by

$$U = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \right\} \quad (8)$$

$$(c) \ C = \left\{ \begin{bmatrix} i \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \end{bmatrix} \right\}$$

Hint: Recall that we define the complex inner product between $\mathbf{a}, \mathbf{b} \in \mathbb{C}^n$ as $\langle \mathbf{a}, \mathbf{b} \rangle = \mathbf{a}^\top \mathbf{b}^$*

Solution:

We start with the vectors of C as our basis, and we want to find an orthonormal basis for this same space using Gram-Schmidt. As with the previous subparts, we define the following for notational convenience:

$$C = \{\mathbf{c}_1, \mathbf{c}_2\}, U = \{\mathbf{u}_1, \mathbf{u}_2\} \quad (9)$$

We summarize the first two steps of the Gram-Schmidt algorithm as follows:

- i. $\mathbf{u}'_1 = \mathbf{c}_1; \quad \mathbf{u}_1 = \frac{\mathbf{u}'_1}{\|\mathbf{u}'_1\|}.$
- ii. $\mathbf{u}'_2 = \mathbf{c}_2 - \langle \mathbf{c}_2, \mathbf{u}_1 \rangle \mathbf{u}_1; \quad \mathbf{u}_2 = \frac{\mathbf{u}'_2}{\|\mathbf{u}'_2\|}.$

For the span of C , this is

i. $\mathbf{u}'_1 = \mathbf{c}_1 = \begin{bmatrix} i \\ 1 \end{bmatrix}$.

Observe that $\|\mathbf{u}'_1\| = \sqrt{\langle \mathbf{u}'_1, \mathbf{u}'_1 \rangle} = \sqrt{(\mathbf{u}'_1)^\top (\mathbf{u}'_1)^*} = \sqrt{(i)(-i) + (1)(1)} = \sqrt{-(-1) + 1} = \sqrt{2}$

Therefore, $\mathbf{u}_1 = \frac{\mathbf{u}'_1}{\|\mathbf{u}'_1\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} i \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$.

ii.

$$\begin{aligned} \mathbf{u}'_2 &= \mathbf{c}_2 - \langle \mathbf{c}_2, \mathbf{u}_1 \rangle \mathbf{u}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \left\langle \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} \frac{i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \right\rangle \begin{bmatrix} \frac{i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}^\top \begin{bmatrix} \frac{i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}^* \right) \begin{bmatrix} \frac{i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}^\top \begin{bmatrix} \frac{-i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \right) \begin{bmatrix} \frac{i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} - \left(\frac{-i}{\sqrt{2}} \right) \begin{bmatrix} \frac{i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} -\frac{1}{2} \\ \frac{i}{2} \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{2} \\ \frac{i}{2} \end{bmatrix} \implies \mathbf{u}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ i \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} \end{bmatrix} \end{aligned}$$

Thus, our new orthonormal basis U is given by

$$U = \left\{ \begin{bmatrix} \frac{i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix}, \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} \end{bmatrix} \right\} \quad (10)$$

3. Functional Gram-Schmidt

In this question, we will be demonstrating an example of the Gram Schmidt process, with functions!

For this question, we will define the inner product between two functions f and g as follows:

$$\langle f(x), g(x) \rangle = \int_{-1}^1 f(x)g(x)dx$$

Intuitively, we can think of the inner product of functions as similar to the vector inner product! Instead of taking the sum of the products of each pair of terms, we are taking the following:

$$f(a)g(a)dx + f(a+dx)g(a+dx)dx + \cdots + f(b)g(b)dx = \int_a^b f(x)g(x)dx, \text{ as } dx \rightarrow 0$$

Note that the norm of a function, $\|f(x)\|$, is defined as $\sqrt{\langle f(x), f(x) \rangle} = \sqrt{\int_{-1}^1 f(x)^2 dx}$.

Construct an **orthonormal** basis that spans $\mathcal{P}_1$, the set of all polynomials of at most degree 1.

Hint: The basis $\{1, x\}$ spans $\mathcal{P}_1$.

Solution: We know that the set $\{1, x\}$ spans $\mathcal{P}_1$, so we will start with that basis.

(a)

$$p_1 = 1$$

This is not normalized, as $\sqrt{\int_{-1}^1 1^2 dx} = \sqrt{2}$, so instead, we have:

$$p_1 = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

(b)

$$p_2 = x - \left\langle \frac{\sqrt{2}}{2}, x \right\rangle \frac{\sqrt{2}}{2} = x - \int_{-1}^1 \frac{\sqrt{2}}{2} x dx = x - 0 = x$$

Note that this is not normalized, as $\|p_2\| = \sqrt{\langle p_2, p_2 \rangle} = \sqrt{\int_{-1}^1 x^2 dx} = \sqrt{\frac{2}{3}}$.

It follows that our unit polynomial is actually:

$$p_2 = \sqrt{\frac{3}{2}} x = \frac{\sqrt{6}}{2} x$$

Our constructed basis is the following set:

$$\left\{ \frac{\sqrt{2}}{2}, \frac{\sqrt{6}}{2} x \right\}$$

4. Correctness of the Gram-Schmidt Algorithm

Suppose we take a list of real vectors $\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$ in $\mathbb{R}^m$ and run the following Gram-Schmidt algorithm on it to perform orthonormalization. It produces the vectors $\{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_n\}$.

Algorithm 1 The Gram-Schmidt Algorithm

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1: for  $i = 1$  up to  $n$  do                                     ▷ Iterate through the vectors
2:    $\mathbf{r}_i = \mathbf{a}_i - \sum_{j < i} (\mathbf{q}_j^\top \mathbf{a}_i) \mathbf{q}_j$                  ▷ Find the amount of  $\mathbf{a}_i$  that remains after we project
3:   if  $\mathbf{r}_i = \mathbf{0}$  then
4:      $\mathbf{q}_i = \mathbf{0}$ 
5:   else
6:      $\mathbf{q}_i = \frac{\mathbf{r}_i}{\|\mathbf{r}_i\|}$                                        ▷ Normalize the vector.
7:   end if
8: end for

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In this problem, we will prove the correctness of the Gram-Schmidt algorithm by showing that the following three properties hold on the vectors output by the algorithm:

- (a) If $\mathbf{q}_i \neq \mathbf{0}$, then $\mathbf{q}_i^\top \mathbf{q}_i = \|\mathbf{q}_i\|^2 = 1$ (i.e. the $\mathbf{q}_i$ have unit norm whenever they are nonzero).
- (b) For all $i \neq j$, $\mathbf{q}_i^\top \mathbf{q}_j = 0$ (i.e. $\mathbf{q}_i$ and $\mathbf{q}_j$ are orthogonal).
- (c) For all $1 \leq \ell \leq n$, $\text{span}(\mathbf{a}_1, \dots, \mathbf{a}_\ell) = \text{span}(\mathbf{q}_1, \dots, \mathbf{q}_\ell)$.

- (a) First, use the construction from the if/then/else statement in the algorithm to show that the first property holds. In particular, show that $\|\mathbf{q}_i\| = 1$ if $\mathbf{q}_i \neq \mathbf{0}$.

Solution:

When $\mathbf{r}_i = \mathbf{0}$, line 4 of the algorithm will make it terminate with $\mathbf{q}_i = \mathbf{0}$. This means that the nonzero case must come from the else statement. Observing line 6 of the algorithm, we have $\mathbf{q}_i = \frac{\mathbf{r}_i}{\|\mathbf{r}_i\|}$. We know the norm of $\|\mathbf{q}_i\| = \mathbf{q}_i^\top \mathbf{q}_i$. Expanding this, we see

$$\|\mathbf{q}_i\|^2 = \mathbf{q}_i^\top \mathbf{q}_i = \left(\frac{\mathbf{r}_i}{\|\mathbf{r}_i\|} \right)^\top \left(\frac{\mathbf{r}_i}{\|\mathbf{r}_i\|} \right). \quad (11)$$

Grouping terms and using the definition of the norm, we get

$$\|\mathbf{q}_i\|^2 = \frac{1}{\|\mathbf{r}_i\|^2} \mathbf{r}_i^\top \mathbf{r}_i = \frac{1}{\|\mathbf{r}_i\|^2} \|\mathbf{r}_i\|^2 = 1. \quad (12)$$

- (b) Next, we want to establish orthogonality between every pair of vectors in $\{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_n\}$. For this stage in the proof, prove that if we know that the first $k-1$ vectors we have constructed, $\{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_{k-1}\}$, are orthogonal, then the newly constructed vector $\mathbf{q}_k$ (from the k th iteration of our algorithm) must also be orthogonal to all of those vectors.

This relationship automatically implies that $\{\mathbf{q}_1, \mathbf{q}_2\}$ are orthogonal to each other, which implies that each distinct pair in $\{\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3\}$ are orthogonal to each other, and so on until we have shown that every pair of distinct vectors in $\{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_n\}$ are orthogonal to each other.

Hint: Notice that the case $\mathbf{q}_k = \mathbf{0}$ is always true, since the inner product of any vector with $\mathbf{q}_k = \mathbf{0}$ is 0. As a result, focus on the case $\mathbf{q}_k \neq \mathbf{0}$ and expand an expression for $\mathbf{q}_k$ using the algorithm specification.

Note: This proof technique is formally known as strong induction, but its specific mechanics and correctness are not within the scope of this class. You do not need to understand it to solve this problem.

Solution:

Per the hint, we know that all that we need to deal with is $\mathbf{q}_k$ itself. We need to verify that it is orthogonal to all the previous $\mathbf{q}_j \in \{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_{k-1}\}$. If $\|\mathbf{r}_k\| = 0$, then $\mathbf{q}_k = \mathbf{0}$ and so it definitely holds since the zero vector is orthogonal to every vector. So now we can assume $\|\mathbf{r}_k\| > 0$.

For any $j < k$,

$$\mathbf{q}_j^\top \mathbf{q}_k = \mathbf{q}_j^\top \frac{1}{\|\mathbf{r}_k\|} \left(\mathbf{a}_k - \sum_{n < k} \mathbf{q}_n (\mathbf{q}_n^\top \mathbf{a}_k) \right) \quad (13)$$

$$= \frac{1}{\|\mathbf{r}_k\|} \left(\mathbf{q}_j^\top \mathbf{a}_k - \mathbf{q}_j^\top \sum_{n < k} \mathbf{q}_n (\mathbf{q}_n^\top \mathbf{a}_k) \right) \quad (14)$$

$$= \frac{1}{\|\mathbf{r}_k\|} \left(\mathbf{q}_j^\top \mathbf{a}_k - \sum_{n < k} \mathbf{q}_j^\top \mathbf{q}_n (\mathbf{q}_n^\top \mathbf{a}_k) \right) \quad (15)$$

$$= \frac{1}{\|\mathbf{r}_k\|} \left(\mathbf{q}_j^\top \mathbf{a}_k - \mathbf{q}_j^\top \mathbf{a}_k \right) = 0 \quad (16)$$

The final step is a cancellation of the cross terms, since they are all zero. Only when the index $n = j$ in the sum, the $\mathbf{q}_j^\top \mathbf{q}_j = 1$ and $\mathbf{q}_j^\top \mathbf{a}_k$ survives.

- (c) Lastly, we show the final property by considering each ℓ from 1 to n , and showing the statement that $\text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_\ell\} = \text{span}\{\mathbf{q}_1, \dots, \mathbf{q}_\ell\}$. This statement is true when $\ell = 1$ since the algorithm produces $\mathbf{q}_1$ as a scaled version of $\mathbf{a}_1$. Now prove that if the spans match for $\ell = k - 1$, then they must also match for $\ell = k$.

This implies that because $\text{span}\{\mathbf{a}_1\} = \text{span}\{\mathbf{q}_1\}$, then so too is $\text{span}\{\mathbf{a}_1, \mathbf{a}_2\} = \text{span}\{\mathbf{q}_1, \mathbf{q}_2\}$, and so forth, until we get that $\text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_n\} = \text{span}\{\mathbf{q}_1, \dots, \mathbf{q}_n\}$. Observe that this is another example of strong induction, like the previous subpart.

Hint: What you need to show is that if there exist $\alpha_1, \alpha_2, \dots, \alpha_k$ such that $\mathbf{y} = \sum_{j=1}^k \alpha_j \mathbf{a}_j$, then there exist $\beta_1, \beta_2, \dots, \beta_k$ such that $\mathbf{y} = \sum_{j=1}^k \beta_j \mathbf{q}_j$. This shows the forward direction; you should also show the reverse direction.

Try writing $\mathbf{a}_k$ in terms of $\mathbf{q}_k$ and earlier $\mathbf{q}_j$, and use the condition for $\ell = k - 1$ to show the condition for $\ell = k$. Don't forget the case that $\mathbf{q}_k = \mathbf{0}$.

Solution:

Proof outline: we first show that we can express β in terms of α , then vice versa. To do so, we express $\mathbf{a}_k$ or $\mathbf{q}_k$ using the algorithm above and then group $\mathbf{q}_j$ terms.

Assume for all $\ell \leq k - 1$, for any $\alpha_1, \dots, \alpha_{k-1}$ there exists $\beta_1, \dots, \beta_{k-1}$ such that

$$\sum_{j=1}^{k-1} \beta_j \mathbf{q}_j = \sum_{j=1}^{k-1} \alpha_j \mathbf{a}_j \quad (17)$$

Then for $\ell = k$, consider a generic set of α 's, $\alpha_1, \dots, \alpha_{k-1}, \alpha_k$.

$$\sum_{j=1}^k \alpha_j \mathbf{a}_j = \alpha_k \mathbf{a}_k + \sum_{j=1}^{k-1} \alpha_j \mathbf{a}_j \quad (18)$$

$$= \alpha_k \left(\|\mathbf{r}_k\| \mathbf{q}_k + \sum_{i < k} \mathbf{q}_i \left(\mathbf{q}_i^\top \mathbf{a}_k \right) \right) + \sum_{j=1}^{k-1} \beta_j \mathbf{q}_j \quad (19)$$

$$= \underbrace{\alpha_k \|\mathbf{r}_k\|}_{\beta_{k'}} \mathbf{q}_k + \sum_{j=1}^{k-1} \left(\underbrace{\beta_j + \alpha_k \left(\mathbf{q}_j^\top \mathbf{a}_k \right)}_{\beta_{j'}} \right) \mathbf{q}_j \quad (20)$$

The second equality is from line 2 of the algorithm where it states $\mathbf{r}_k = \mathbf{a}_k - \sum_{i < k} \mathbf{q}_i \left(\mathbf{q}_i^\top \mathbf{a}_k \right)$. This can be rearranged as

$$\mathbf{a}_k = \mathbf{r}_k + \sum_{i < k} \mathbf{q}_i \left(\mathbf{q}_i^\top \mathbf{a}_k \right) \quad (21)$$

This finishes the first direction of the proof, as we can now write the vector $\mathbf{y}$ as

$$\mathbf{y} = \sum_{j=1}^k \beta_{j'} \mathbf{q}_j. \quad (22)$$

Thus, it holds for $\ell = k$. The opposite direction follows similarly, which is shown below.

Once again, we assume that the statement holds for $\ell = k - 1$. For any $\beta_1, \dots, \beta_{k-1}$ there exists $\alpha_1, \dots, \alpha_{k-1}$ such that

$$\sum_{j=1}^{k-1} \beta_j \mathbf{q}_j = \sum_{j=1}^{k-1} \alpha_j \mathbf{a}_j \quad (23)$$

Then for $\ell = k$, consider a generic set of β 's, $\beta_1, \dots, \beta_{k-1}$.

$$\mathbf{y} = \sum_{j=1}^k \beta_j \mathbf{q}_j = \beta_k \mathbf{q}_k + \sum_{j=1}^{k-1} \beta_j \mathbf{q}_j \quad (24)$$

$$= \beta_k \frac{\mathbf{r}_k}{\|\mathbf{r}_k\|} + \sum_{j=1}^{k-1} \beta_j \mathbf{q}_j \quad (25)$$

$$= \frac{\beta_k}{\|\mathbf{r}_k\|} \left(\mathbf{a}_k - \sum_{j < k} \mathbf{q}_j \left(\mathbf{q}_j^\top \mathbf{a}_k \right) \right) + \sum_{j=1}^{k-1} \beta_j \mathbf{q}_j \quad (26)$$

$$= \frac{\beta_k}{\|\mathbf{r}_k\|} \mathbf{a}_k + \sum_{j=1}^{k-1} \mathbf{q}_j \left(\beta_j - \frac{\beta_k}{\|\mathbf{r}_k\|} \left(\mathbf{q}_j^\top \mathbf{a}_k \right) \right). \quad (27)$$

Notice that the second sum is only up through $k - 1$ and involves a linear combination of the $\mathbf{q}_j$. So we can use our assumption: there exists an appropriate set of α that would weight $\mathbf{a}_j$ to equal that second sum.

$$\mathbf{y} = \frac{\beta_k}{\|\mathbf{r}_k\|} \mathbf{a}_k + \sum_{j=1}^{k-1} \alpha_j \mathbf{a}_j \quad (28)$$

Here, α_k can be defined to be $\frac{\beta_k}{\|\mathbf{r}_k\|}$ and we have proved the result for $\ell = k$. Thus, we have the new form of

$$\mathbf{y} = \sum_{j=1}^k \alpha_j \mathbf{a}_j. \quad (29)$$

5. QR Proofs

- (a) Let $\mathbf{A} = \mathbf{QR}$, where $\mathbf{Q}$ is an orthonormal matrix and $\mathbf{R}$ is an upper triangular matrix. $\mathbf{A}$, $\mathbf{Q}$, and $\mathbf{R}$ are all $n \times n$ matrices. Show that if $\mathbf{A}$ is invertible, then $\mathbf{R}$ is also invertible.

Solution:

Let $\mathbf{A} = [\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_n]$ and $\mathbf{Q} = [\mathbf{q}_1 \ \mathbf{q}_2 \ \cdots \ \mathbf{q}_n]$

$$\mathbf{A} = \mathbf{QR} = [\mathbf{q}_1 \ \mathbf{q}_2 \ \cdots \ \mathbf{q}_n] \begin{bmatrix} \|\mathbf{z}_1\| & \langle \mathbf{q}_1, \mathbf{a}_2 \rangle & \langle \mathbf{q}_1, \mathbf{a}_3 \rangle & \cdots \\ 0 & \|\mathbf{z}_2\| & \langle \mathbf{q}_2, \mathbf{a}_3 \rangle & \cdots \\ 0 & 0 & \|\mathbf{z}_3\| & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{bmatrix}$$

Note that $\mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3, \dots, \mathbf{z}_n$ are the orthogonalized (but not normalized) versions of $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3, \dots, \mathbf{a}_n$ respectively. Also note that $\mathbf{z}_1 = \mathbf{a}_1$. Recall from lecture that an upper-triangular matrix $\mathbf{R}$ is invertible if and only if all of its diagonal entries are non-zero. Thus, we have to show that if $\mathbf{A}$ is invertible, then $\|\mathbf{z}_i\| \neq 0$ for all $i = 1, 2, \dots, n$. Expanding $\mathbf{A} = \mathbf{QR}$, we get

$$\begin{aligned} \mathbf{a}_1 &= \|\mathbf{z}_1\| \mathbf{q}_1 \\ \mathbf{a}_2 &= \langle \mathbf{q}_1, \mathbf{a}_2 \rangle \mathbf{q}_1 + \|\mathbf{z}_2\| \mathbf{q}_2 \\ \mathbf{a}_3 &= \langle \mathbf{q}_1, \mathbf{a}_3 \rangle \mathbf{q}_1 + \langle \mathbf{q}_2, \mathbf{a}_3 \rangle \mathbf{q}_2 + \|\mathbf{z}_3\| \mathbf{q}_3 \\ &\vdots \\ \mathbf{a}_n &= \langle \mathbf{q}_1, \mathbf{a}_n \rangle \mathbf{q}_1 + \langle \mathbf{q}_2, \mathbf{a}_n \rangle \mathbf{q}_2 + \cdots + \langle \mathbf{q}_{n-1}, \mathbf{a}_n \rangle \mathbf{q}_{n-1} + \|\mathbf{z}_n\| \mathbf{q}_n \end{aligned}$$

Let us attempt a proof by contradiction. Assume that if $\mathbf{A}$ is invertible, then $\mathbf{R}$ is not invertible. This implies that there exists some i such that $\|\mathbf{z}_i\| = 0$ (i.e., some element along the diagonal of $\mathbf{R}$ that is 0). This implies that $\mathbf{z}_i = \mathbf{a}_i - \sum_{j=1}^{i-1} \langle \mathbf{q}_j, \mathbf{a}_i \rangle \mathbf{q}_j = \mathbf{0}$. Recall that if we encounter the zero vector while performing Gram-Schmidt, that must mean that our original set of vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$ are linearly dependent. This contradicts the fact that $\mathbf{A}$ is invertible. Therefore, our initial assumption that $\mathbf{R}$ is not invertible is false, and we have shown that if $\mathbf{A}$ is invertible, then $\mathbf{R}$ is also invertible.

Note: It is true in general that if $\mathbf{A}$ has full rank (not necessarily square), then $\mathbf{R}$ is invertible.

- (b) Let $\mathbf{A} = \mathbf{QR}$, where $\mathbf{A}$ has linearly independent columns, $\mathbf{Q}$ is an orthonormal matrix, and $\mathbf{R}$ is an upper triangular matrix. Show that $\mathbf{A}$ and $\mathbf{Q}$ have the same column space, i.e., the span of the columns of $\mathbf{A}$ and $\mathbf{Q}$ are the same.

Solution: Let $\mathbf{A} = [\mathbf{a}_1 \ \mathbf{a}_2 \ \cdots \ \mathbf{a}_n]$ and $\mathbf{Q} = [\mathbf{q}_1 \ \mathbf{q}_2 \ \cdots \ \mathbf{q}_n]$. To show that $\text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\} = \text{span}\{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_n\}$, we need to show that any vector in $\text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$ is also in $\text{span}\{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_n\}$, and vice versa.

- i. If $\mathbf{b} \in \text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$, then there exists some $\mathbf{x}$, such that $\mathbf{Ax} = \mathbf{b}$

$$\mathbf{Ax} = \mathbf{QRx} = \mathbf{Q(Rx)} = \mathbf{b}$$

Let $\mathbf{y} = \mathbf{Rx}$. Then $\mathbf{Qy} = \mathbf{b}$. Therefore, $\mathbf{b} \in \text{span}\{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_n\}$.

- ii. If $\mathbf{b} \in \text{span}\{\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_n\}$, then there exists some $\mathbf{x}$, such that $\mathbf{Qx} = \mathbf{b}$. We know that since $\mathbf{A}$ has linearly independent columns, $\mathbf{R}^{-1}$ exists, such that $\mathbf{Q} = \mathbf{AR}^{-1}$

$$\mathbf{Qx} = \mathbf{AR}^{-1}\mathbf{x} = \mathbf{b}$$

Let $\mathbf{y} = \mathbf{R}^{-1}\mathbf{x}$. Then $\mathbf{Ay} = \mathbf{b}$. Therefore, $\mathbf{b} \in \text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$.

6. A Cute Decomposition

In this course we have learned about the QR Decomposition, in which you take a matrix A and decompose it into two matrices Q and R , where Q is orthonormal and R is upper triangular. For this problem, we would like to derive the "Cute" decomposition, such that

$$A = QT$$

where Q is still an orthonormal matrix, but T is now a **lower triangular** matrix.

- (a) Given a matrix A , describe in words a process by which you could find the "Cute" decomposition.

Hint: Think about how we find the QR decomposition for a matrix. How does this change with the "Cute" decomposition?

Solution: We recall that the QR decomposition appears as follows:

$$\begin{bmatrix} | & | & \cdots & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \\ | & | & \cdots & | \end{bmatrix} = \begin{bmatrix} | & | & \cdots & | \\ \mathbf{q}_1 & \mathbf{q}_2 & \cdots & \mathbf{q}_n \\ | & | & \cdots & | \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ 0 & r_{22} & \cdots & r_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & r_{nn} \end{bmatrix}$$

Where $\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_n$ can be derived from the Gram-Schmidt Algorithm. When finding the QR decomposition, we begin Gram-Schmidt from the first column vector $\mathbf{a}_1$, because we see in the above multiplication that

$$\mathbf{a}_1 = r_{11}\mathbf{q}_1$$

i.e. the first vector $\mathbf{q}_1$ is simply a scaled version of the original vector $\mathbf{a}_1$. Our "Cute" Decomposition, however, would appear as follows:

$$\begin{bmatrix} | & | & \cdots & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \\ | & | & \cdots & | \end{bmatrix} = \begin{bmatrix} | & | & \cdots & | \\ \mathbf{q}_1 & \mathbf{q}_2 & \cdots & \mathbf{q}_n \\ | & | & \cdots & | \end{bmatrix} \begin{bmatrix} t_{11} & 0 & \cdots & 0 \\ t_{21} & t_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ t_{n1} & t_{n2} & \cdots & t_{nn} \end{bmatrix}$$

In this decomposition, we can see that now

$$\mathbf{a}_n = t_{nn}\mathbf{q}_n$$

the last vector $\mathbf{q}_1$ is a scaled version of the last original vector $\mathbf{a}_1$. Therefore, we can find the "Cute" decomposition in the same manner as the QR decomposition, but by doing Gram-Schmidt in reverse order!

- (b) Using the scheme you devised in the previous subpart, find the "Cute" decomposition of the following matrix A :

$$A = \begin{bmatrix} 2 & 0 & 1 \\ -3 & 3 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

Solution: As we figured out in the previous subpart, the "Cute" Decomposition requires us to perform Gram-Schmidt on the provided matrix, but starting from the last vector. In other words:

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{a}_2 = \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix}, \quad \mathbf{a}_3 = \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix}$$

Now we perform Gram-Schmidt to find $\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3$:

$$\mathbf{q}_1 = \frac{1}{\|\mathbf{a}_1\|} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}$$

$$\begin{aligned} \mathbf{r}_2 &= \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix} - \langle \mathbf{a}_2, \mathbf{q}_1 \rangle \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix} - \sqrt{3} \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \mathbf{q}_2 &= \frac{1}{\|\mathbf{r}_2\|} \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix} \\ &= \frac{1}{\sqrt{6}} \begin{bmatrix} -1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \end{bmatrix} \end{aligned}$$

$$\begin{aligned} \mathbf{r}_3 &= \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix} - \langle \mathbf{a}_3, \mathbf{q}_1 \rangle \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix} - \langle \mathbf{a}_3, \mathbf{q}_2 \rangle \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix} - 0 + \frac{9}{\sqrt{6}} \begin{bmatrix} -\frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ -3 \\ 1 \end{bmatrix} + \begin{bmatrix} -\frac{3}{2} \\ 3 \\ -\frac{3}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ 0 \\ -\frac{1}{2} \end{bmatrix} \end{aligned}$$

$$\mathbf{q}_3 = \frac{1}{\|\mathbf{r}_3\|} \begin{bmatrix} \frac{1}{2} \\ 0 \\ -\frac{1}{2} \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$$

Putting our vectors together, we can see that

$$Q = \begin{bmatrix} | & | & | \\ \mathbf{q}_3 & \mathbf{q}_2 & \mathbf{q}_1 \\ | & | & | \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix}$$

From here, because we know that $A = QT$ and that Q is orthonormal, it must be the case that

$$T = Q^{-1}A = Q^T A$$

Therefore,

$$T = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{bmatrix} \begin{bmatrix} 2 & 0 & 1 \\ -3 & 3 & 1 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & 0 \\ -\frac{3\sqrt{3}}{\sqrt{2}} & \sqrt{6} & 0 \\ 0 & \sqrt{3} & \sqrt{3} \end{bmatrix}$$

As a result, the "Cute" Decomposition of our matrix is given by:

$$Q = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix}, T = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & 0 \\ -\frac{3\sqrt{3}}{\sqrt{2}} & \sqrt{6} & 0 \\ 0 & \sqrt{3} & \sqrt{3} \end{bmatrix}$$

7. Homework Process and Study Group

Whom did you work with on this homework? List names and student ID's. (In case you met people at homework party or in office hours, you can also just describe the group.) How did you work on this homework? If you worked in your study group, explain what role each student played for the meetings this week.

Solution:

I first worked by myself for 2 hours, but got stuck on problem 5. Then I met with my study group.

XYZ played the role of facilitator ... etc. We were still stuck on problem 5 so we went to office hours to talk about the problem.

Then I went to homework party for a few hours, where I finished the homework.